

SOME FURTHER RESULTS CONCERNING REGRESSION TO THE MEAN¹

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Chinn, S. and R. F. Heller (Dept. Community Medicine, St. Thomas' Hospital Medical School, London SE1 7EH, England). Some further results concerning regression to the mean. *Am J Epidemiol* 1981;114:902-905.

Algebraic formulae are derived for the regression to the mean effect for a measurement whose population mean and variance change over time and for subpopulations selected according to initial value in any defined range. The distribution of values for the whole population is assumed to be normal. The results are applied to plasma cholesterol levels of participants in a dietary intervention study, some of whom were selected as having high initial levels. The formulae enable the effects of intervention to be separated from secular change and regression to the mean in studies for which this was not previously possible.

cholesterol; population; risk; comparative study; statistics

The concept of regression to the mean has been well explained by Healy and Goldstein (1) but problems still arise over quantification of the effect when studies are made of groups of individuals selected as having, for example, a high blood pressure or high cholesterol level. James (2) and Davis (3) gave the formulae for the mean and variance of a variable at a second occasion for subjects selected as having a measurement above a defined value at the first occasion. However, they both made the assumption that the mean and variance of the unselected population remained stable. Blood pressure and cholesterol increase with age, and hence time, so that this assumption is unjustified. We present the corresponding formulae for variables with changing mean and variance, and for any defined interval of initial values of the variable used to select a

subpopulation of subjects. We apply the results to a study in which subjects were selected for dietary advice on a number of criteria including raised plasma cholesterol.

METHOD

We denote the variate corresponding to initial values of the variable of interest by X and that corresponding to values at the subsequent occasion by Y . We assume that X is normally distributed mean μ_x , variance σ_x^2 , over the whole population from which subjects are selected, and that similarly Y is normally distributed mean μ_y , variance σ_y^2 . In addition, we assume that the correlation between X and Y is ρ .

Consider a subpopulation selected as having values x of X satisfying $k_1 < x < k_2$. For example, if we selected subjects with systolic blood pressure in the range 220 mmHg to 240 mmHg, k_1 would be 220 and k_2 240 mmHg. When $k_2 = \infty$ this corresponds to the problem considered by James (2) and Davis (3), all subjects with high values of X , greater than k_1 , being selected.

Let $z_1 = (k_1 - \mu_x)/\sigma_x$, $z_2 = (k_2 - \mu_x)/\sigma_x$, the subpopulation limits on the stan-

Received for publication December 29, 1980, and in final form June 3, 1981.

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The authors thank Professor G. Rose for permission to use data from the United Kingdom Heart Disease Prevention Project.

standardized scale for the whole population, and define

$$\eta = - \left\{ \phi(z_2) - \phi(z_1) \right\} / \left\{ \Phi(z_2) - \Phi(z_1) \right\}$$

$$\lambda = - \left\{ z_2 \phi(z_2) - z_1 \phi(z_1) \right\} / \left\{ \Phi(z_2) - \Phi(z_1) \right\}$$

where

$$\phi(u) = 1/\sqrt{2\pi} \exp(-\frac{1}{2}u^2),$$

$$\Phi(u) = \int_{-\infty}^u \phi(t) dt.$$

When $k_2 = \infty$, $\eta = c_1$ as defined by Davis (or k_0 as defined by James) and $\lambda = c_1 z_1$.

The subpopulation means and variances of X , Y , and $Y-X$ were derived in terms of the above parameters using methods similar to those given by James (2).

RESULTS

The algebraic expressions for the means and variances of X , Y and $Y-X$ for a subpopulation selected so that $k_1 < x < k_2$ are given in table 1. When $k_2 = \infty$, $\sigma_y = \sigma_x$ and $\mu_y = \mu_x$ the results reduce to those given by Davis and James.

APPLICATION

In a coronary heart disease prevention trial (4), 24 factories in England and Wales were randomly assigned to either an intervention group or a control group. All men aged 40 to 59 years in the intervention factories, and a 10 per cent sample of men in the same age group in the control factories, were invited for a car-

diovascular screening examination which included estimation of plasma cholesterol from a capillary blood sample. Each man was assigned a score based on his age, blood pressure, smoking habits and plasma cholesterol level; a higher score was given to each succeeding interval on the cholesterol scale, the intervals being: less than 210 mg/100 ml, 210-220 mg/100 ml, . . . 250-260 mg/100 ml or greater than 260 mg/100 ml.

By this risk score 10 to 15 per cent of subjects in each intervention factory were designated as at high risk and given intensive dietary advice. Four years later their plasma cholesterol levels were re-measured as part of an annual assessment of risk factors. Initial and four-year plasma cholesterol measurements on 669 high risk subjects from the intervention factories were available for analysis.

Measurements of plasma cholesterol at the same time points were available for 475 subjects from the control factories and from these the following parameters were estimated:

$$\hat{\mu}_x = 217.0 \text{ mg/100 ml,}$$

$$\hat{\mu}_y = 226.5 \text{ mg/100 ml,}$$

$$\hat{\sigma}_x = 38.2 \text{ mg/100 ml,}$$

$$\hat{\sigma}_y = 38.4 \text{ mg/100 ml,}$$

$$\hat{\rho} = 0.69$$

Although the distributions of cholesterol levels in the control group were slightly positively skewed they were better approximated by the normal distribution than the two-parameter log-normal, and so calculations were made on this basis.

TABLE 1

Means and variances of the initial value (X), final value (Y) and change (Y-X) for a subpopulation selected according to the initial value of X, $k_1 < x < k_2$

Variate	Mean	Variance
X	$\mu_x + \eta\sigma_x$	$(1 + \lambda - \eta^2)\sigma_x^2$
Y	$\mu_y + \rho\eta\sigma_y$	$\{1 + \rho^2(\lambda - \eta^2)\}\sigma_y^2$
$Y-X$	$\mu_y - \mu_x + \eta(\rho\sigma_y - \sigma_x)$	$(1 + \lambda - \eta^2)\sigma_x^2 + \{1 + \rho^2(\lambda - \eta^2)\}\sigma_y^2 - 2\rho(1 + \lambda - \eta^2)\sigma_x\sigma_y$
Covariance (X,Y)		$\rho\sigma_x\sigma_y(1 + \lambda - \eta^2)$

TABLE 2

Expected and observed means and standard deviations of initial and final cholesterol levels for control and intervention groups, by initial level

	Initial plasma cholesterol level (mg/100 ml)						
	<210	210-	220-	230-	240-	250-	260-
Observed data for control subjects							
Initial mean	183.2	214.6	224.0	234.7	244.4	255.0	280.4
SD	22.0	2.5	2.9	3.2	3.0	2.7	20.4
Final mean	201.7	224.6	226.0	239.5	253.8	260.8	270.6
SD	29.9	22.0	27.9	25.5	23.1	34.8	31.4
Expected values derived from formulae in table 1							
Initial mean	181.9	215.0	225.0	234.9	244.8	254.8	279.1
SD	21.8	2.9	2.9	2.9	2.9	2.9	16.4
Final mean	202.2	225.1	232.0	238.9	245.8	252.7	269.6
SD	31.6	27.9	27.9	27.9	27.9	27.9	30.0
Observed data for intervention subjects							
Initial mean	177.6	214.6	224.3	234.2	244.1	254.0	286.0
SD	32.6	2.9	2.5	2.8	2.7	2.8	30.2
Final mean	196.0	217.1	218.8	235.5	235.1	241.2	255.2
SD	30.4	22.9	27.0	34.7	28.6	31.8	37.6
No. of subjects							
Control	203	60	50	37	32	30	63
Intervention	42	20	58	47	94	102	306
η	-0.9181	-0.0521	0.2082	0.4685	0.7288	0.9891	1.6267
λ	0.1682	-0.9916	-0.9509	-0.7748	-0.4631	-0.0160	1.8311

The means and standard deviations calculated from the formulae in table 1 are given in table 2, below the observed data with which they are in good agreement. The values of η and λ used in the calculations are also given. Only for the two subpopulations with the highest initial levels does the expected regression to the mean effect outweigh the expected increase in cholesterol with age, shown by the difference between the expected initial and final mean values. The observed values for the intervention group are also shown, the initial values confirming their comparability with the control group. The effect of the trial's intervention activities (5) is clearly shown by the differences between the observed final cholesterol levels for the intervention group and those expected.

DISCUSSION

When a control group is followed up, as in the example given, direct estimates of the required parameters can be obtained and the assumptions of normality tested. In theory the control subjects can be divided on their initial measurement using the same criteria by which the intervention subjects were selected and direct comparisons made of the corresponding intervention and control groups, without the need for the results reported here. However, in practice the size of the subgroups so formed may be small, and provided the assumption of normality is justified a better comparison will be obtained with the use of the formulae given in this paper. As variance is estimated from the whole group, provided only that the total

number of subjects is large, the error in the estimates for table 1 will be small compared to the variation in the observed subpopulation means; standard errors of the means can be calculated from the estimated variances in the usual way to provide a rough and ready test of statistical significance.

When no control subjects are followed up either the assumption of a stable mean and variance must be made so that the procedure described by James (2) can be used or estimates of μ_y , σ_y and ρ obtained from elsewhere. The instability of the population mean and variance may be due entirely to changes with age. In this case, provided subjects of a sufficiently wide age range are included in the initial survey, μ_y , σ_y can be estimated from the initial measurements, in addition to μ_x and σ_x . An estimate of ρ can then be obtained by equating the observed covariance of the initial and final measurements of the intervention group with

the expression given in table 1, or where several subpopulations were used, to the appropriate average, substituting the estimates of σ_x and σ_y .

Clearly no procedure can allow for secular trends or changes in measurement procedures if no control subjects are followed up. Estimates of the regression to the mean effect are now available for all studies of reasonable design.

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